$$= \lim_{h\to 0}\frac{(x+h)^n-(x)^n}{h}$$

$$= \lim_{h\to 0}\frac{(x+h)^n-(x)^n}{(x+h)-x}$$

$$= \lim_{z\to x}\frac{z^n-x^n}{z-x}, \text{ where } z=x+h \text{ and } z\to x \text{ as } h\to 0$$

$$= nx^{n-1} \qquad \left[\because \lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}\right]$$

i.e., $\frac{d}{dx}(x^n) = nx^{n-1}$

2. $\frac{d}{dx}(e^x) = e^x$

Proof:

Let $f(x) = e^x$

$\therefore f(x+h) = e^{x+h}$

$$\frac{d}{dx}(f(x)) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{e^{x+h}-e^x}{h}$$

$$= \lim_{h\to 0}\frac{e^x e^h-e^x}{h}$$

$$= \lim_{h\to 0} e^x\left[\frac{e^h-1}{h}\right]$$

$$= e^x \lim_{h\to 0}\left[\frac{e^h-1}{h}\right]$$

$$= e^x\times 1 \quad \left[\because \lim_{x\to 0}\left[\frac{e^x-1}{x}\right]=1\right]$$

$$= e^x$$

i.e., $\frac{d}{dx}(e^x) = e^x$.

3. $\frac{d}{dx}(\log_e x) = \frac{1}{x}$

Proof:

Let $f(x) = \log_e x$

$\therefore f(x+h) = \log_e(x+h)$

$$\frac{d}{dx}(f(x)) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{\log_e(x+h)-\log_e x}{h}$$

$$= \lim_{h\to 0}\frac{\log_e\left(1+\frac{h}{x}\right)}{h}$$

$$= \lim_{h\to 0}\frac{\log_e\left(1+\frac{h}{x}\right)}{\frac{h}{x}}\ \frac{1}{x}$$

$$= \frac{1}{x}\lim_{h\to 0}\frac{\log_e\left(1+\frac{h}{x}\right)}{\frac{h}{x}}$$

$$= \frac{1}{x}\times 1$$

$$\left[\because \lim_{x\to 0}\frac{\log_e(1+x)}{x}=1\right]$$

$\frac{d}{dx}(\log_e x) = \frac{1}{x}$

Example 5.31

Find $\frac{d}{dx}(x^3)$ from first principle.

Solution

Let $f(x) = x^3$

$\therefore\ f(x+h) = (x+h)^3$

$$\frac{d}{dx}(f(x)) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$= \lim_{h\to 0}\frac{(x+h)^3-x^3}{h}$$

$$= \lim_{h\to 0}\frac{3x^2h+3xh^2+h^3}{h}$$

$$= \lim_{h\to 0}(3x^2+3xh+h^2)$$

$$= 3x^2+3x(0)+(0)^2$$

$$= 3x^2$$

$\frac{d}{dx}(x^3) = 3x^2$

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